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LLM Learning Path - Level 2

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Building Steps for LLMs

01

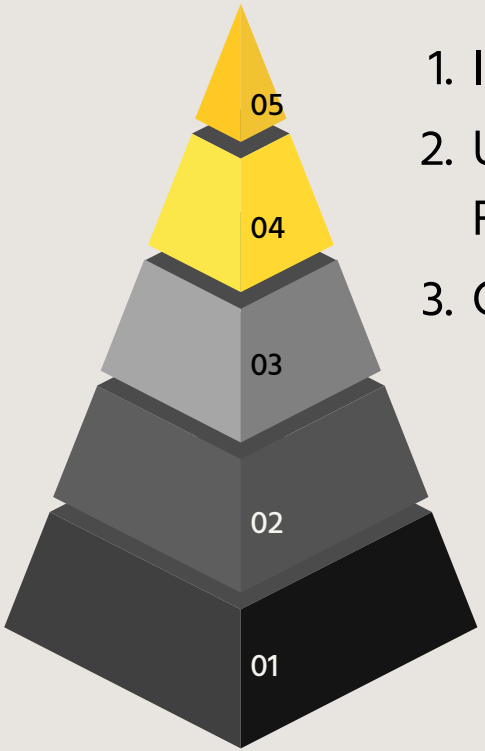
Foundation

Powerful language models trained on extensive text data, forming the basis for various language tasks.

02

DataPrep

Converting documents into instruction pairs, like QA pairs, facilitating fine-tuning and tasks.



Contents at a Glance

1. Introduction to Language Models
2. Understanding LLM Architecture / Foundation Models
3. Getting Started with LLM Data Studio
   - Clean Data for Reliable NLP Models
   - Examples of data preparation for LLM downstream tasks
   - Effortless Data Prep with LLM DataStudio
   - LLM DataStudio Supported Workflows
   - Generate your own dataset
   - The Workflow Builder
   - Preparation of a Question Answering Dataset

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Essential key functions in data preparation for LLMs

1. Data Object
2. Data Augmentation
3. Text Cleaning
4. Profanity Check
5. Text Quality Check
6. Length Checker
7. Valid Question
8. Pad Sequence
9. Truncate Sequence by Score
10. Compression Ratio Filter
11. Boundary Marking
12. Sensitive Info Checker
13. RLHF Protection
14. Language Understanding
15. Data Deduplication
16. Toxicity Detection
17. Output

Curating Data for LLM Tasks:

Extract Key Information: Pick out the significant facts from the article, such as types of exercises, health impacts, and challenges.

Create Q&A Pairs: Transform the key points into questions and provide the corresponding answers based on the article's content.

Curating Data for LLM Tasks:

Examples:

Q: What are the different types of exercises discussed in the article?
A: The article covers aerobic, strength training, and flexibility exercises.

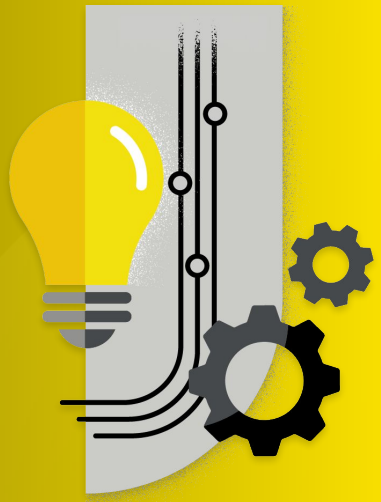
Q: How does exercise influence overall health?
A: Engaging in regular exercise has been shown to improve cardiovascular health, boost mood, and enhance physical fitness.

Q: What challenges might people face when starting an exercise routine?
A: Some challenges include lack of motivation, time constraints, and the need for proper guidance.

Enhancing LLM Data with LLM DataStudio

LLM DataStudio features:

- Q&A Generative of text and audio data
- Text Cleaning
- Data Quality Issue Detection
- Tokenization
- Text Length Control



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LLM DataStudio Supported Workflows

1. **Question and Answer Workflow:**

- ❑ Preparing Datasets for Question Answering Models
- ❑ Structured Datasets with Context, Questions, and Answers
- ❑ Crucial for Accurate User Query Responses

2. **Text Summarization Workflow:**

- ❑ Handling Articles and Summaries
- ❑ Extracting Key Information for Concise Summaries
- ❑ Training Summarization Models for Informative Summaries

3. **Instruct Tuning Workflow:**

- ❑ Creating Datasets with Prompts and Responses
- ❑ Training Models to Understand and Follow Instructions
- ❑ Effective Responses to User Prompts

4. **Human - Bot Conversations Workflow:**

- ❑ Organizing Dialogues between Humans and Chatbots
- ❑ Enhancing Conversational Model Training
- ❑ Understanding User Intents and Providing Contextual Responses

5. **Continued PreTraining Workflow:**

- ❑ Preparing Extensive Text Datasets for Pretraining
- ❑ Organizing Long Texts for Enhanced Language Models
- ❑ Improving Language Understanding and Generation

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Clean Data for Reliable NLP Models

- Text Classification
- Named Entity Recognition (NER)
- Text Summarization
- Sentiment Analysis
- Question Answering
- Machine Translation
- Text Generation
- Text Completion
- Text Segmentation
- Natural Language Understanding (NLU)
- Natural Language Generation (NLG)

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Structured Data Preparation Workflow in LLM DataStudio

LLM DataStudio follows a structured data preparation process.

The process includes several stages:

- ❑ Data intake
- ❑ Workflow construction
- ❑ Configuration
- ❑ Assessment
- ❑ Result generation

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Importance of Clean Data in Downstream NLP Tasks

- ➔ Improved Model Performance
- ➔ Mitigated Bias and Unwanted Influences
- ➔ Consistency and Coherence
- ➔ Enhanced Generalization
- ➔ Ethical Considerations
- ➔ Improved User Experience and Trust

The Workflow Builder

1. Create Workflow:
- Add Processing Steps
- Select from Available Options
- Arrange in Desired Order

2. Run and Save:
- After Workflow Definition
- Click "RUN" to Save Progress
- Proceed to Configuration Page

3. Clear Workflow:
- Start Fresh or Modify
- Click "CLEAR" to Reset Canvas

4. Delete Steps:
- Remove Specific Steps
- Right-Click Step
- Select Delete Option

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Workflow Builder Tool

Key Attributes:

- Drag and Drop: ***Easy Addition of Preparation Steps***
- Linear Pipeline: ***Ensures Smooth Flow***
- Customization: ***Fine-Tune Processing***
- Input and Output: ***Configurable Columns and Formats***

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Configuring Datasets for Question Answering Workflow

1. Question Column:

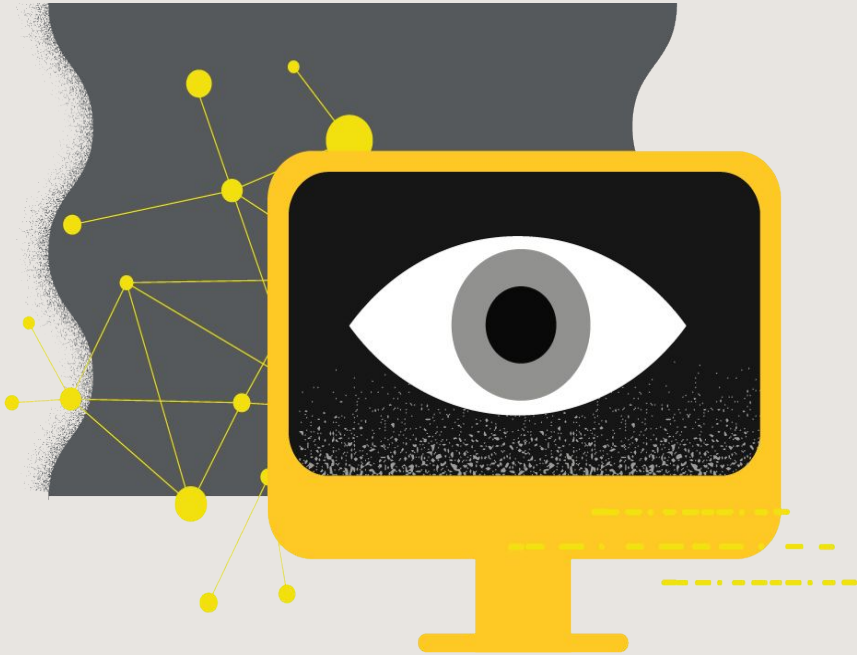
- Specify the Column Containing Questions
- Designate as the "Question Column"

2. Answer Column:

- Indicate the Column with Corresponding Answers
- Set as the "Answer Column"

3. Context Column:

- Identify Column with Additional Information
- Related to Questions and Answers
- Assign as the "Context Column"



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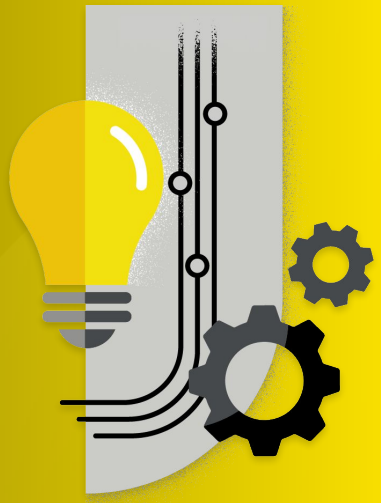
Workflow Builder Activities

- **Create Workflow:**
  - Users arrange processing steps on the canvas from available options.
- **Run and Save:**
  - Click "RUN" to save and proceed to configuration after defining the workflow.
- **Clear Workflow:**
  - Click "CLEAR" to reset the canvas for a fresh start or edits.
- **Delete Steps:**
  - Remove steps by right-clicking and selecting delete.



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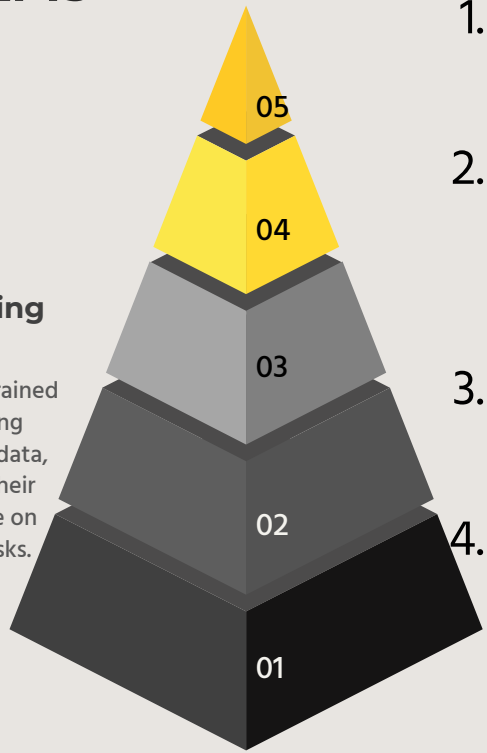
Fine-tuning

Refining pre-trained models using task-specific data, enhancing their performance on targeted tasks.



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4. Fine-tuning LLMs
   - Fine-tuning Process and Techniques
   - LLM Studio for fine tuning
   - Deploy to Hugging Face

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Fine-Tuning Large Language Models (LLMs)

Key Subjects:

- LLM Fine-Tuning Techniques Reminder
- Task-specific Data Importance
- Selecting Model Backbones
- Deep Dive into Fine-Tuning Process
- Quantisation and LoRA Techniques
- Optimizing Large Language Models
- Using LLM Studio for Fine Tuning
- Deploying Models to HuggingFace

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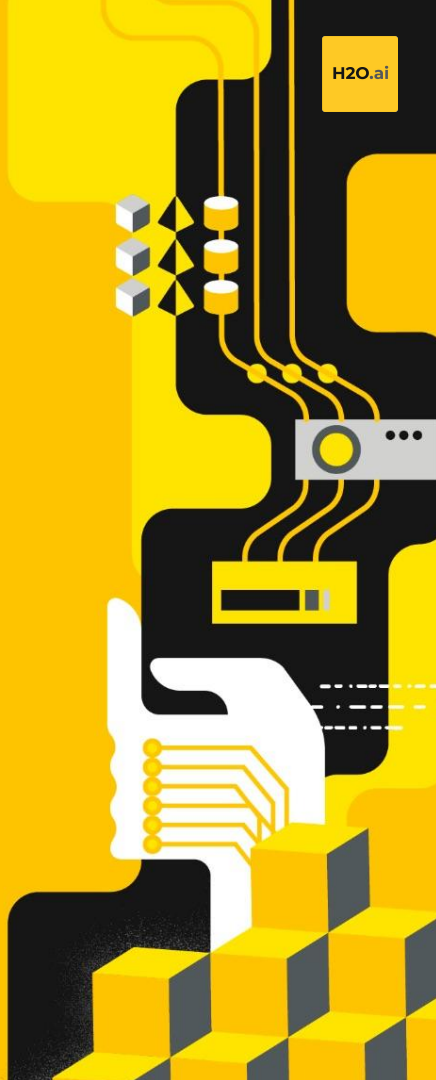
H2O.ai:

- is a strong advocate for open-source initiatives.
- is committed to supporting data-related efforts that benefit community knowledge.
- aims to enhance user experiences through its support for open-source projects.
- promotes accessibility in data-related initiatives.
- encourages open-source collaboration as part of its core values.



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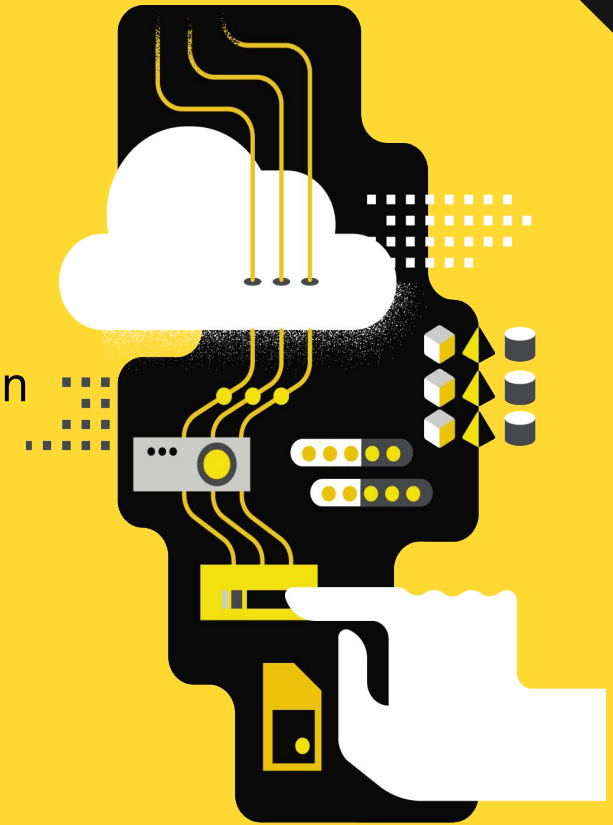
Fine-tuning tailors a pre-trained language model to specific tasks.

Why Fine-Tune?

- **Specialization**: Fine-tuning tailors LLMs for specific tasks.
- **Data Efficiency**: Reduces data requirements by leveraging pre-existing knowledge.
- **Faster Development**: Accelerates NLP application creation.
- **Cost Savings**: More cost-effective than training from scratch.
- **Transfer Learning**: Applies prior knowledge to boost task performance.
- **Continuous Learning**: LLMs adapt for diverse applications.



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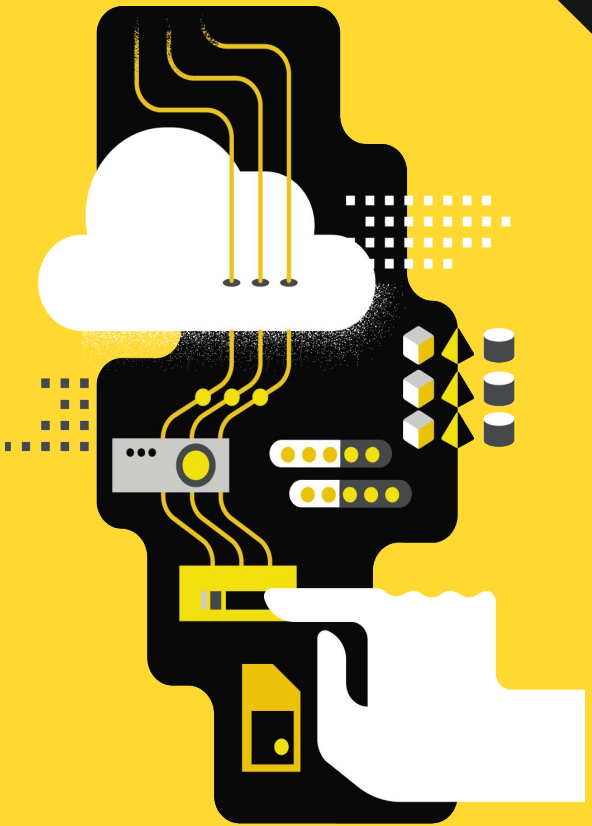
What are Backbones?

- They refer to the foundational architecture and training data.
- Backbones form the core structure and knowledge base.
- They offer the fundamental understanding and language capabilities supporting the broader LLM ecosystem.
- Backbones are the basis on which various language-related applications and capabilities are built.



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Factors to consider in choosing Backbones

Key Differentiators for Backbones:

- ❑ Model Size
- ❑ Number of Parameters

Performance vs. Training Time:

- ❑ Larger Models: Better Performance
- ❑ Trade-off: Longer Training Duration

Practical Approach:

- ❑ Start with a Smaller Model
- ❑ If Desired Performance Not Met, consider Upgrading to a Larger Model

What are Synthetic datasets?

- Synthetic datasets are artificially created datasets that mimic real-world data without being derived from actual observations.

- These datasets are typically generated through algorithms, simulations, or generative models to simulate patterns, structures, and features similar to genuine data.

- They are valuable in situations where obtaining authentic data is challenging, costly, or restricted.

- Synthetic datasets can effectively replace real data in various applications, including machine learning, data analysis, and testing.

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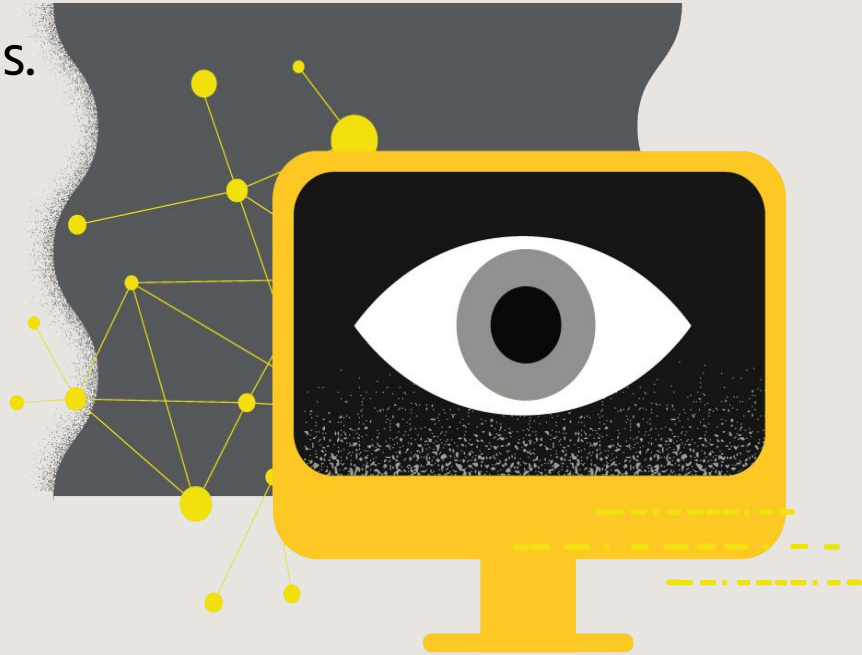
Key Aspects regarding Synthetic Datasets

- **Data Generation**: Creating synthetic data involves using rules and models to mimic real-world data characteristics.
- **Controlled Experiments**: Synthetic datasets offer precise control over experiment parameters, enabling accurate hypothesis testing and algorithm evaluation.
- **Privacy and Security**: Synthetic data is a safe way to share information without revealing personal data.
- **Data Augmentation**: Synthetic data supplements real data, increasing training data for better machine learning model performance.
- **Validation and Testing**: Synthetic datasets are useful for testing applications when real data is scarce, offering controlled testing environments.

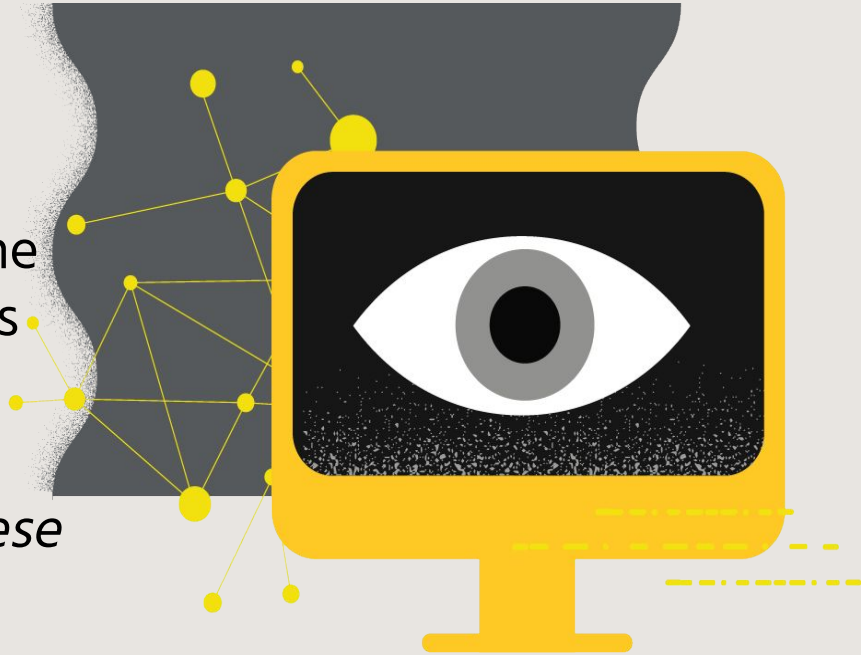
Synthetic images are valuable for:

- Training image recognition algorithms.
- Evaluating algorithm performance.
- Enabling rigorous testing.
- Supporting algorithm refinement.



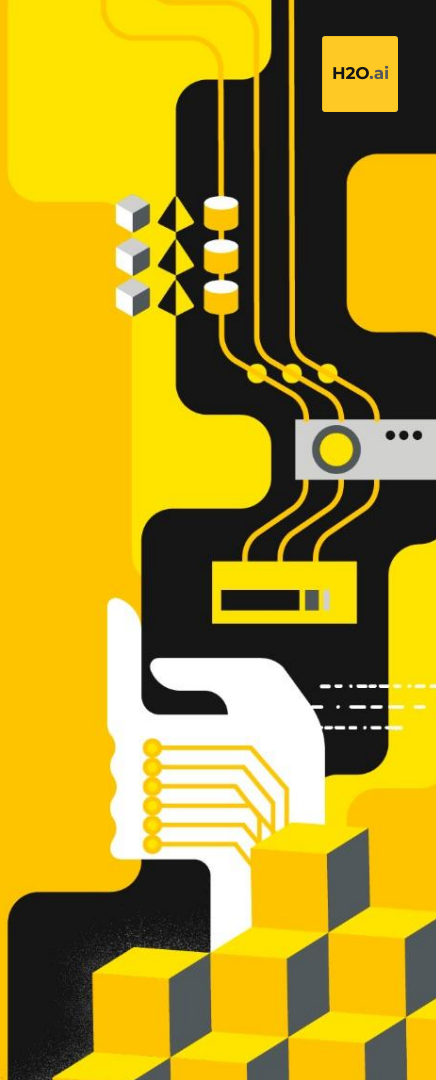
Synthetic data has its own set of limitations:

- They may not replicate all the intricate details of real-world data.
- The quality of synthetic data relies on the accuracy of the models and assumptions used in their creation.



Researchers should be cautious about these limitations when incorporating synthetic data into their applications.

- **Relevance**: The dataset should align closely with the LLM's intended task, such as using medical records for medical diagnosis predictions.
- **Bias & Fairness**: Preventing biases in the dataset is crucial to avoid unfair or harmful model predictions.
- **Quality**: Thorough data cleaning is vital, as a single bad example can significantly impact the model's performance.



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Key Factors Influencing Fine-Tuning Success

- The quality of fine-tuning hinges on the dataset it relies upon.
- To achieve the desired performance in the target task:
  - Prioritize data relevance
  - Ensure data diversity
  - Strive for unbiased data
  - Maintain thorough data annotation

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H2O.ai's fine-tuned h2oGPT models:

1. Mitigate risks tied to advanced language models, including bias, privacy, and copyright issues.

2. Promote accessibility, transparency, and fairness through open-source Large Language Models (LLMs).

3. Widen AI access and ensure equitable distribution of AI benefits.

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Here's how backbones aid in fine-tuning:

- **Transfer Learning**: Pre-trained backbones reduce data and time requirements.
- **Domain Adaptation**: They adapt to specialized domains.
- **Parameter Efficiency**: Modify only a fraction of parameters.
- **Resource Savings**: Faster and more efficient than training from scratch.
- **Improved Performance**: Enhance model performance for specific tasks.

To select the right backbone for fine-tuning, consider these tips:

- Understand your task and its nuances.
- Match model architecture to task requirements.
- Assess model size and resource compatibility.
- Evaluate data quality and quantity.
- Align with the task's domain.
- Consider multilingual capabilities if necessary.
- Ensure hardware supports the chosen model.
- Check model performance on benchmarks.
- Seek community support and documentation.
- Be open to experimentation and adapt based on results.

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Quantization

- Involves reducing the precision of numerical values.
- Replaces high-precision values (e.g., 32-bit floating-point) with lower bit-width representations (e.g., 8-bit or lower).
- Aims to optimize memory and computation efficiency in neural networks.

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Quantization serves two primary purposes:

1. **Reduced Model Size:**
   - Fewer bits for numerical values make models smaller.
   - Ideal for resource-constrained devices and lowers storage needs.

2. **Faster Inference:**
   - Lower-precision values lead to quicker inference.
   - Critical for real-time applications like mobile devices and edge computing.

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LoRA (Low-Rank Adaptation)

- Compresses neural networks by reducing matrix ranks.

- This lowers parameter count, leading to more efficient models.

- Benefits include reduced memory usage and faster inference.

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Quantization *involves decreasing numerical precision in neural networks to enhance efficiency.*

LoRA *reduces the rank of specific weight matrices for model compression and optimization.*

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Deploying your model on H2O LLM Studio provides several advantages, including:

- Increased reach for sharing
- Simplified integration
- The opportunity to receive valuable feedback
- Contributing to the advancement of AI

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Key Insights to Remember

1. Customizing LLMs for specific tasks is pivotal, offering efficiency, savings, and adaptability.

2. H2O LLM Studio streamlines LLM fine-tuning without coding, providing real-time insights.

3. Synthetic datasets mimic real-world data when real data is limited.

4. Choosing the right LLM backbone is crucial for specific tasks.

5. LLM optimization improves efficiency and scalability.

6. Quantization and LoRA boost LLM efficiency.

7. We demonstrated H2O LLM Studio and model deployment for hands-on learning.

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Thank you!